

Kaelin Wulf

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Education

Reed College

August 2020 – May 2024

B.A., Neuroscience; Cumulative GPA: 3.51

Commendation for Excellence in Scholarship - 2022 & 2023

Thesis: [Glia Cell Dynamics in the Zebrafish Optic Tectum](#)

Research Experience

Post-Baccalaureate Microscopy Engineer

June 2024 - present

Allen Institute for Neural Dynamics; Advisor: Adam Glaser

- Developed, characterized, aligned and operated custom large-scale light-sheet microscopes for imaging whole expanded mouse brains.
- Authored an imaging [protocol](#) and detailed documentation [wiki](#) on GitHub; and drafted a manuscript on the microscope and properties of expansion microscopy (in-progress).
- Improved optical and mechanical designs through CAD modeling, machining, and prototyping novel optical subassemblies.
- Led the assembly of multiple microscope systems and independently performed alignment, optimization, and troubleshooting.

Senior Thesis Student

August 2023 - May 2024

Reed College Neuroscience Department; Advisor: Kara Cervený

- Conducted a yearlong independent thesis research project studying the cell dynamics and patterns of microglia and oligodendrocyte progenitor cells (OPCs) in the optic tectum of larval zebrafish.
- Optimized an image acquisition and analysis pipeline to work with large data on limited hardware and trained team on best imaging practices.
- Produced a ~100-page written thesis and passed oral defense.

Undergraduate Research Fellow

May - August 2023

Reed College Neuroscience Department; Advisor: Kara Cervený

- Led a summer research project on the topic of cell death and microglia cell dynamics in the optic tectum of larval zebrafish.
- Generated transgenic crosses and single-cell injectable plasmid vectors, and imaged larvae using confocal and light-sheet microscopy.

Undergraduate Research Fellow

May - August 2022

Reed College Neuroscience Department; Advisors: Erik Zornik & Charlotte Barkan

- Proposed, managed, and executed a collaborative research project investigating androgen-dependent neuroanatomy in *Xenopus laevis* vocal-motor circuits.
- Conducted animal surgeries, dissections, and histological preparations including IHC and tissue clearing for light-sheet imaging.

Student Intern

June - August 2020

Air Force Research Laboratory; Advisor: Wellesley Pereira

- Collected and analyzed photometric telescope images of satellites to develop an approach to extracting fundamental physical properties of space objects.
- Collaborated with remote teams under virtual limitations to design novel methods for multispectral data analysis.

Leadership Experience

Confidential Advocate and Coordinator

September 2022 – May 2024

Reed College SHARE program

- Provided confidential advocacy and support for survivors of interpersonal violence.
- Facilitated weekly full team meetings, and led working groups focused on education and reshaping campus culture around topics of sexual violence.
- Organized community events and led workshops on consent, supporting survivors, and the neurobiology of trauma.

Honors/Awards

Summer Undergraduate Research Fellowship 2022 & 2023

The Beckman Foundation, and the Helen Stafford Summer Research Fellowship Fund

Undergraduate Research Opportunity Grant 2022 & 2023

The Galakatos Science Research Fund, and Paul K. Richter & Evalyn Elizabeth Cook Richter Memorial Fund

Initiative Grant in Undergraduate Research 2024

Reed College, Neuroscience Department

Microscopy Image Competition 2024

Northwest Developmental Biology Conference, Nikon Exhibition

Publications

Wulf, K., Jiang, X., Glaser, A., et al. “An optimized expansion-assisted selective plane illumination microscope for nanoscale imaging of centimeter scale tissues”. (in preparation)

Wulf, K. and Nguyen, A.; Yue, J., Green, K., and Cervený, K. “Tectal microglia show altered developmental dynamics in optic nerve degeneration zebrafish models”. (in preparation)

Conference Presentations and Proceedings

Wulf, K., Jiang, X., Woodard, M., Mwaniki, W., Glaser, A. “[Optimized expansion-assisted selective plane illumination microscopy \(ExA-SPIM\)](#)”. *SPIE Photonics West*, 2026. Paper 13859-8. [Abstract & Talk]

Cao, K., Flannery, R., Javeri, R., **Wulf, K.**, Grotz, P., Arshadi, C., Ouellette, N., Hooper, M., Woodard, M., Jiang, X., Friedmann, D., Roy, L., Summers, M., Tasic, B., Svoboda, K., Glaser, A., Chandrashekar, J. “Organization of the thalamic reticular nucleus across spatial scales” *Society for Neuroscience*, 2025.

Wulf, K. and Javeri, R.; Vasquez, S., Arshadi, C., Ouellette, N., Jiang, X., Baka, J., Kovacs, G., Woodard, M., Seshamani, S., Cao, K., Clack, N., Grim, A., Turschak, E., Liddell, A., Rohde, J., Grotz, P., Logsdon, M., Feng, D., Svoboda, K., Chandrashekar, J., and Glaser, A. “[New technologies for mapping centimeter scale tissues with nanoscale resolution](#)” *Lake Conference* 2025, [Poster].

Cao, K., Flannery, R., Javeri, R., **Wulf, K.**, Grotz, P., Arshadi, C., Ouellette, N., Hooper, M., Woodard, M., Jiang, X., Friedmann, D., Roy, L., Summers, M., Tasic, B., Svoboda, K., Glaser, A., Chandrashekar, J. “Multiscale molecular anatomy of the thalamic reticular nucleus” *Lake Conference* 2025.

Wulf, K., Jiang, X., and Glaser, A. “[Characterization of a new expansion-assisted selective plane illumination microscope](#)”. *Optica Biophotonics Congress* 2025, Technical Digest Series (Optica Publishing Group, 2025), paper DM3A.2. [Abstract & Talk]

Jiang, X., **Wulf, K.**, Woodard, M., Mwaniki, W., Glaser, A. “A new expansion-assisted selective plane illumination microscope for nanoscale imaging of centimeter-scale tissues”. *Optica Biophotonics Congress 2025*, Technical Digest Series (Optica Publishing Group, 2025), paper NM5C.2.

Wulf, K., Jiang, X., and Glaser, A. “[Expansion-assisted selective plane illumination microscopy for nanoscale imaging of centimeter-scale tissues](#)”. *O-STEM*, 2024. [Abstract & Poster]

Wulf K. and Cervený K. “[Effects of innervation on oligodendrocyte and microglia cell dynamics in the zebrafish visual system](#)”. *Society for Developmental Biology, Northwest Regional Meeting*, 2024. [Poster]

Wulf K., Yue J., and Cervený K. “[Dynamics of cell death in the denervated zebrafish optic tectum](#)”. Faculty for Undergraduate Neuroscience; *Society for Neuroscience*, 2023. [Poster]

Wulf, K., Truong, B.B., Zornik E., and Barkan C. “[Examining Androgen Induced Changes to Vocal Neurons of Female *Xenopus Laevis*](#)”. Faculty for Undergraduate Neuroscience; *Society for Neuroscience*, 2022. [Poster]

Professional Associations

Society for Neuroscience (SFN) 2022-present

Society of Photo-Optical Instrumentation Engineers (SPIE) 2025-present

Optica 2024-present

Skills

Engineering: Optical alignment, Basic CAD design, Soldering, Manual machining, Testing and characterizing optics, Specifying/purchasing Parts, Mechanical construction

Wet Lab: Animal surgery/dissection, PCR, Genotyping, Plasmid cloning, Microinjections, Tissue sectioning, Immunolabeling, Tissue clearing

Software: Python, R, Photoshop/Illustrator, 3D Image Analysis (Imaris), ImageJ/FIJI